

Double Fault: Artifacts Produced by Physical Activity Can Impact the Efficacy of Chronic Local Field Potential Recordings and Adaptive Stimulation

Marjolein Muller, MSc^{1,2}  and M. Fiorella Contarino, MD, PhD^{1,2,*} 

We greatly appreciated the recent report by Sousa & Tinkhauser (2025),¹ which highlighted the issue of cardiac-induced artifacts on local field potentials (LFP) recorded from implantable deep brain stimulation (DBS) systems. Their reported patient with Parkinson's disease (PD), had requested to have the neurostimulator implanted in the left chest to facilitate his right-handed forehand in tennis. The authors demonstrate how these artifacts can affect adaptive stimulation (aDBS) and how they can be identified during in-office recordings.

We report here a right-handed gentleman with PD (age around 70), also a tennis player, who underwent bilateral subthalamic nucleus (STN) DBS with SenSight (B33005) leads due to treatment-resistant tremor, and slight bradykinesia and rigidity. The Medtronic Percept™ PC stimulator (Minneapolis, MN) was implanted in the right chest. After a standard monopolar review, stimulation was set on contact points 2 and 10, at 180 Hz, 60us, and 1.7 mA and 0.7 mA, respectively. Initial recordings in the BrainSense™ SetUp mode showed a clear beta peak (17.58 Hz) for the right STN, highest for contact pair 9–11, and a less evident beta peak (19.53 Hz) in the left STN, highest for contact pair 1–3, which were selected for at-home recordings (Fig. 1A,B). No ECG artifacts were detected by online inspection and with the offline Perceive Toolbox (Matlab R2022b).²

About 6 weeks postoperatively, the patient obtained a substantial reduction of tremor, rigidity, and bradykinesia, and improvement of fatigue. This allowed him to resume playing tennis, which he had given up due to his symptoms.

Despite reporting no motor fluctuations anymore, on some days but not on others (Fig. 1C–F), chronic recordings showed substantial bilateral increases in the selected beta power, lasting

approximately 2–3 hours. When asked, he reported playing tennis during those days and times. We interpret this bilateral increase in beta power as a movement-related artifact, particularly due to the swinging motion of his right forehand movements.^{3,4}

Our observations add to the considerations reported by Sousa and Tinkhauser,¹ showing that while, as previously reported, left-sided implants may be more prone to ECG artifacts,² right-sided implants could be more likely affected by movement artifacts in right-handed subjects. Movement artifacts resulting in an apparent sustained increase in the beta power, can be mistaken for prolonged off periods, thus erroneously driving overstimulation during aDBS protocols, both in a single-threshold and in a double-threshold mode.

These two cases show how one patients' tennis playing, requiring a left-implanted neurostimulator, possibly contributed to the higher ECG contamination, while the other patients' tennis forehand movements produced significant movement artifacts, which altogether, using tennis jargon, could be named a “double fault.”

Chronic LFP recordings from implantable DBS systems allow collecting potentially valuable information on disease-related brain activity even in the at-home environment. With the recent introduction of aDBS, many DBS centers worldwide have promptly begun integrating this new feature into clinical practice.⁵ We consider our practical observations a valuable warning to always approach brain sensing and aDBS critically and never to neglect patients' feedback in the interpretation of the signals.

¹Department of Neurology, Haga Teaching Hospital, The Hague, The Netherlands; ²Department of Neurology, Leiden University Medical Center, Leiden, The Netherlands

*Correspondence to: Dr. M. Fiorella Contarino, Department of Neurology, Leiden University Medical Centre, Albinusdreef 2, 2333 ZA Leiden, the Netherlands; E-mail: m.f.contarino@lumc.nl

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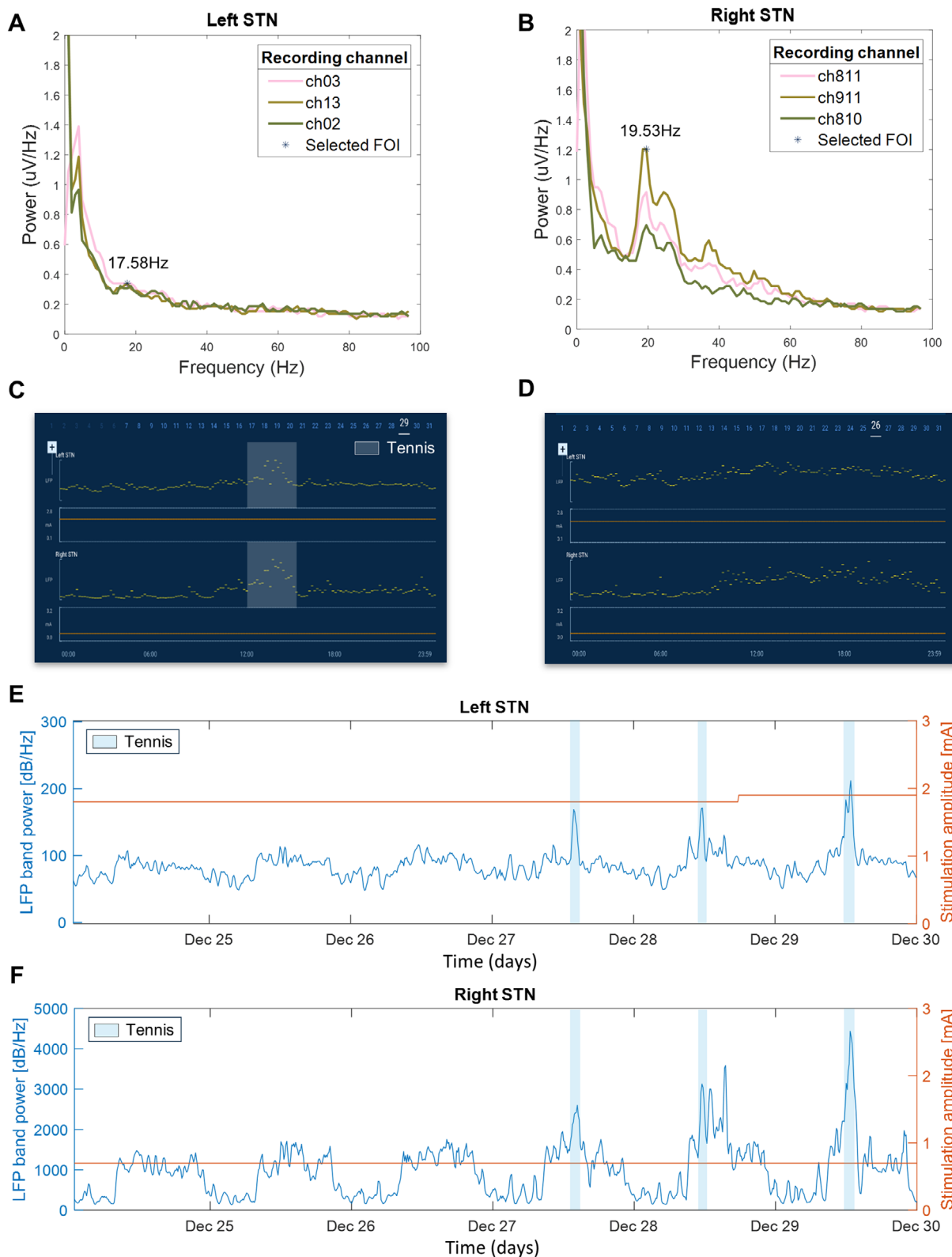


Figure 1. Beta power increase during physical activity (playing tennis). In-office and at-home local field potential recordings performed in a right-handed patient with Parkinson's Disease implanted with a neurostimulator in the right chest. In the left (A) and (Figure legend continues on next page.)

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right (B) hemisphere BrainSense™ SetUp (bipolar) recordings were performed OFF-medication during monopolar review to allow the selection of the frequency of interest (FOI) for at-home recordings (L: 17.58 Hz, recording channel 1–3; R: 19.53 Hz, recording channel 9–11). At-home recordings reviewed in the clinic 6 weeks later showed clear and sustained increases of beta-activity lasting 2–3 h at a moment the patient reported playing tennis (December 29th) (C), while this was not the case on a day (December 26th) the patient did not play tennis (D). Upon further offline review of at-home recordings in the left (E) and right (F) hemispheres, these increases in beta-activity were bilateral, occurred on multiple days (December 27th, 28th and 29th), when the patient reported playing tennis (highlighted in blue), around noon, and were not observed on the days he did not play tennis (December 24th, 25th or 26th).

Author Roles

(1) Research project: A. Conception, B. Organization, C. Execution; (2) Statistical Analysis: A. Design, B. Execution, C. Review and Critique; (3) Manuscript Preparation: A. Writing of the first draft, B. Review and Critique.

M.M.: 2B, 1B, 1C, 3B.

M.F.C.: 1A, 3A, 2A, 2C.

Disclosures

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Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions. ■

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